

Contiguous Allocation

1 Basic Idea

Contiguous Allocation

In contiguous allocation, each file is stored as one continuous run of disk blocks. A file's disk location can be described using only the starting block and the number of blocks in the file.

- The disk is divided into fixed-size blocks.
- A file occupies consecutive blocks, with no gaps between its blocks.
- Example: with 1-KB blocks, a 50-KB file needs 50 consecutive blocks.
- Example: with 2-KB blocks, a 50-KB file needs 25 consecutive blocks.
- If the file does not exactly fill its last block, the unused part of that block is wasted.

2 Figure Example

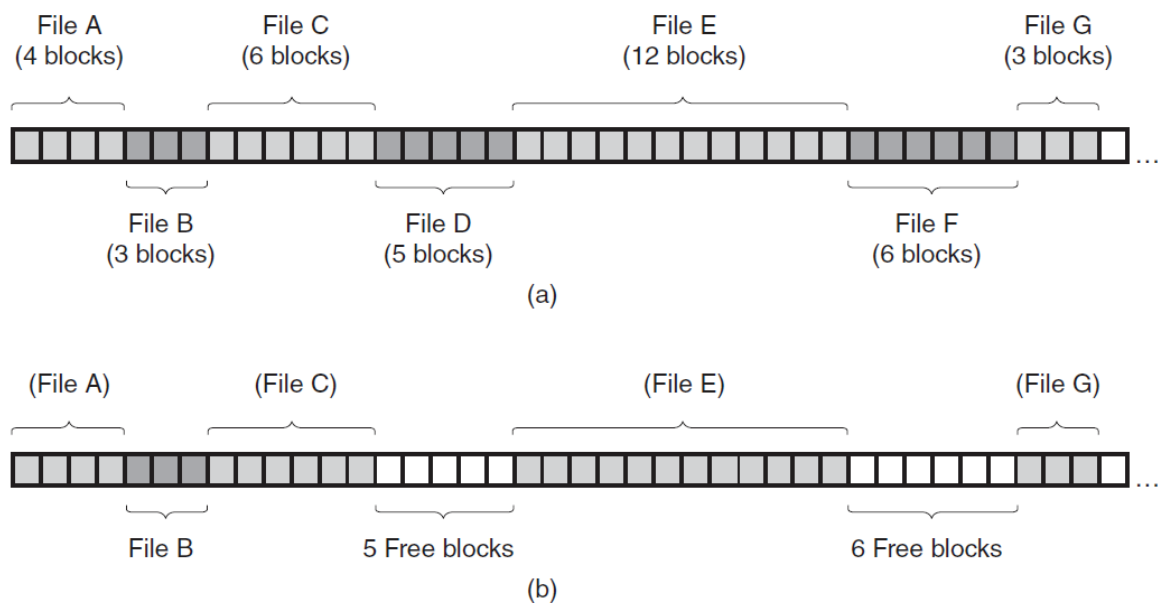


Figure 4-10. (a) Contiguous allocation of disk space for seven files. (b) The state of the disk after files *D* and *F* have been removed.

Figure 1: Contiguous allocation before and after file deletion.

- In part (a), files are stored one after another in consecutive disk blocks.
- File A uses 4 blocks.

- File B uses 3 blocks.
- File C uses 6 blocks.
- File D uses 5 blocks.
- File E uses 12 blocks.
- File F uses 6 blocks.
- File G uses 3 blocks.
- In part (b), files D and F have been removed.
- Their deleted regions become holes of 5 free blocks and 6 free blocks.

3 How the System Tracks a File

Metadata Needed

For each file, the file system only needs:

starting block number + file length in blocks

- If the first block is block b , then the next block is $b + 1$.
- The i th block of the file is found by simple addition.
- This makes the allocation method very easy to implement.

4 Advantages

Advantage	Explanation
Simple metadata	Only the first block and file length must be stored.
Fast lookup	Any file block can be found by addition.
High read speed	The whole file can be read sequentially from disk.
Few seeks	Only one seek is needed to reach the first block.
Good bandwidth	After the first block is reached, data can be read at disk bandwidth.

5 Main Problem: Fragmentation

External Fragmentation

Over time, contiguous allocation creates holes between files. These holes may be free, but they may not be large enough for a new file that needs one long continuous run of blocks.

- When a file is deleted, its blocks become free.
- The disk is not compacted immediately because moving many later blocks is too expensive.

- After many create and delete operations, the disk contains a mixture of files and holes.
- New files must fit into a hole large enough to hold the whole file contiguously.
- This creates external fragmentation.

6 Why File Size Is a Problem

- To choose a suitable hole, the system needs to know the final size of the file.
- This is unrealistic for many files because users and programs often do not know the final size in advance.
- If the estimate is too small, the file may run out of space.
- If the estimate is too large, the system may fail to find a large enough hole.
- Compacting the disk to remove holes can require copying huge amounts of data.

7 Where Contiguous Allocation Works Well

Best Use Case

Contiguous allocation works well when file sizes are known in advance and files do not change after being written.

- It is suitable for CD-ROMs because all file sizes are known before the disk is created.
- It is also useful for write-once optical media such as DVDs and Blu-ray disks.
- DVD movie data may be stored as several contiguous pieces called **extents**.
- An extent is one contiguous physical part of a larger logical file.

8 Comparison

Point	Contiguous Allocation
Storage rule	Store each file in consecutive disk blocks.
Metadata	Starting block and file length.
Performance	Excellent for sequential reads.
Main cost	External fragmentation.
Hard question	How large will the file become?
Good fit	Read-only or write-once media.
Poor fit	General-purpose disks with growing and deleting files.

Final Point

Contiguous allocation is simple and fast, but it is hard to use for changing file systems because deleted files create holes and new files require one large enough continuous free region.